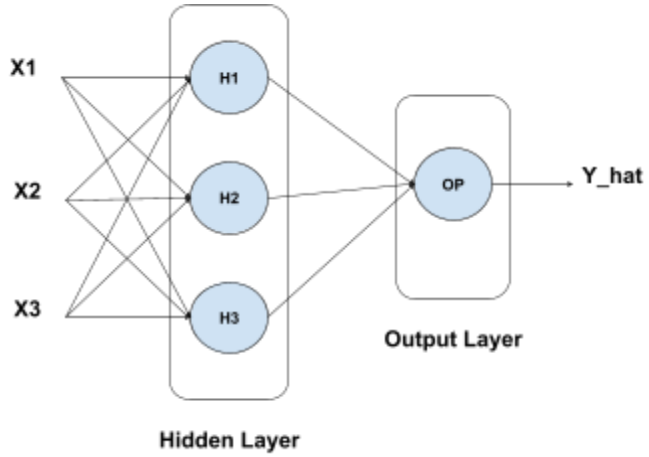
MLP Coursework- Deep Learning Module

Problem 1

Check the following diagram



Let's define the weights like this

- From input to hidden layer, weights are named like Wij, where i is the index of the input node, and j is the index of the hidden node. So weights for H! Will be W11, W21 and W31. The bias is W01.

- From the hidden layer to the output weights are named like Wi_op, where i is the index of the node in the hidden layer. So the weight from H1 to OP is W1op, the bias term for OP is W0op.

Assume that all activation functions are sigmoid, and the given ground truth is represented by **y**

1. Fill in all the weights in this network
2. How many parameters do we have in the network
3. Write the forward transformation equation for all the nodes in the hidden layer and output layer
4. Compute the gradient for OP weights and H3 weights, assuming the loss function to be MSE



Problem 2

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1. Is this ANN suitable for multiclass classification? explain your answer.
2. How will you change your formula of 1.4 if we have 2 nodes in the output layer
3. How will you change your formula of 1.4 if instead of MSE we use binary cross-entropy loss [hint: find the derivative of binary cross-entropy loss]